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Transformers in Artificial Intelligence

Architecture, Evolution, and What They're Actually Used For

A survey of developments from 2017 through early 2026 — covering the core architecture, how it has evolved, what it is being used for, where it falls short, and what comes next.

Abstract

In 2017, a team at Google published a paper called "Attention Is All You Need." The architecture they described — the Transformer — has since become the foundation of nearly every major AI system in use today. GPT, BERT, AlphaFold, DALL-E, the models that write code and read medical scans: all of them trace their lineage to that paper. This survey covers what the Transformer is, why it works so well, how it has evolved, and what it is being used for across medicine, drug discovery, climate science, and beyond.

We also examine where it struggles. The core attention mechanism becomes computationally expensive as sequences grow long — a problem that has driven a wave of alternative architectures in 2023–2025, notably the Mamba family of State Space Models. We assess whether those alternatives are likely to challenge the Transformer's dominance or simply find their own niches alongside it.

The picture that emerges is of a technology still very much in motion. The Transformer is not a fixed invention but a design space that researchers have been refining for nearly a decade — and that process is far from over.

1. Introduction

Before 2017, the dominant approach to sequence modeling in machine learning was the recurrent neural network — specifically the LSTM (Long Short-Term Memory). LSTMs processed text one token at a time, maintaining a hidden state that was supposed to carry information forward across long sequences. In practice, they struggled with long-range dependencies and were slow to train because each step depended on the previous one. Parallelization across a sequence was not possible.

The Transformer threw out the recurrence entirely. Instead of processing tokens one at a time, it compared every token in a sequence to every other token simultaneously — a mechanism called self-attention. The result was more parallelizable, better at capturing relationships between distant words, and, it turned out, dramatically more scalable. Feed it more data and more compute, and performance improves in predictable, measurable ways.

That last property — predictable scaling — is what really changed the field. Once researchers established that bigger models trained on more data reliably produce better results, the race to scale began. GPT-3, released in 2020 with 175 billion parameters, demonstrated that language models trained at sufficient scale could answer questions, write essays, debug code, and pass professional examinations without any task-specific training. This was surprising. Nothing in the original Transformer paper predicted it.

Since then the architecture has been modified extensively — often quietly, through changes in normalization, position encoding, activation functions, and attention mechanisms that collectively make modern models far more efficient and capable than the 2017 original. This paper traces that evolution and explains what each change was trying to fix.

2. How the Transformer Actually Works

2.1 The core idea: attention

The central operation in a Transformer is self-attention. For each token in a sequence, the model computes three vectors: a Query, a Key, and a Value — each derived by multiplying the token's embedding by a learned matrix. To compute the attention score between two tokens, you take the dot product of one's Query with the other's Key, scale it, and pass it through a softmax function to get a probability. The output for each token is then a weighted sum of all Value vectors, where the weights are those probabilities.

What this means in practice: every token is asking, "which other tokens are relevant to understanding me?" A word like "bank" in "river bank" will attend heavily to "river"; the same word in "bank account" will attend to "account." The model learns these relationships from data, not from rules.

> **The attention formula**
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> Attention(Q, K, V) = softmax(QK^T / √d_k) · V Q = Query, K = Key, V = Value, d_k = dimension of the key vectors. The division by √d_k prevents the dot products from growing large in high dimensions, which would push the softmax into regions where gradients nearly vanish.

Multi-head attention runs this process several times in parallel, with each "head" learning different types of relationships — one might track syntactic dependencies, another might track coreference, another might capture semantic similarity. The outputs are concatenated and projected.

The feed-forward network that follows attention in each layer is simpler: two linear transformations with a non-linearity between them, applied identically to each position. It acts as a position-wise memory, storing and transforming representations after attention has gathered context.

2.2 Encoder, decoder, and the variants that took over

The original architecture combined an encoder — which reads the full input at once — with a decoder that generates output one token at a time, attending to both previous output tokens and the encoder's representation. This design was ideal for translation, the original application.

Two simpler variants proved more influential. Encoder-only models like BERT process the full input bidirectionally, making them well-suited for classification and understanding tasks. They were pre-trained by masking random words and asking the model to predict them — forcing it to build deep contextual representations. BERT set state-of-the-art results on eleven benchmarks when it was released in 2018.

Decoder-only models like the GPT series process tokens left-to-right, predicting the next token at each step. This makes them natural text generators. Pre-trained on internet-scale text, they learn a remarkably general model of language that can be directed toward almost any task through prompting alone. GPT-3 in 2020 was the first demonstration of this at a scale that surprised even its creators.

2.3 A decade of architectural improvements

The Transformer that powers a modern large language model looks quite different from the 2017 original, even if the core principles are the same. The main changes:

Pre-normalization. The original design applied layer normalization after each sub-layer. Applying it before (pre-norm) stabilizes gradient flow in very deep networks and is now universal in large models.

RMSNorm. A simpler variant that normalizes only the magnitude of activation vectors rather than centering them. Faster than full layer normalization, with no meaningful accuracy cost.

Rotary Position Embeddings (RoPE). The original model injected position information through fixed sinusoidal encodings added to inputs. RoPE instead encodes position by rotating the Query and Key vectors, which encodes relative position directly in the attention scores. Modern models generalize better to sequence lengths longer than those seen during training.

SwiGLU activation. A gating mechanism in the feed-forward layers — the output of one linear transformation gates another. Consistently improves model quality at equivalent parameter count, and is now standard in LLaMA, Mistral, Gemma, and others.

Grouped-Query Attention (GQA) and Multi-Head Latent Attention (MLA). During inference, a Transformer must cache Key and Value vectors for all previous tokens — the KV cache. This grows linearly with sequence length and can dominate memory at long contexts. GQA groups multiple query heads to share a single KV pair. MLA, introduced by DeepSeek, projects KVs into a lower-dimensional latent space, achieving even larger savings with better accuracy than GQA.

Innovation	What it fixes
Pre-normalization	Unstable gradients in deep networks during training
RMSNorm	LayerNorm overhead — normalizes magnitude only, no centering needed
RoPE position embeddings	Poor length generalization; encodes relative position in attention scores
SwiGLU activation	Suboptimal feed-forward expressivity; gating improves quality at same cost
Grouped-Query Attention	KV cache memory — groups heads to share key/value pairs
Multi-Head Latent Attention	KV cache further; low-rank projection with better

Innovation	What it fixes
	accuracy than GQA
Mixture of Experts	Compute cost of scaling — sparse routing keeps FLOPs manageable
Flash Attention	GPU memory bottleneck in attention — tiled computation, linear memory

Table 1 — Key architectural changes in modern Transformers and what each one addressed

2.4 Mixture of Experts: scaling without proportional cost

One of the most consequential architectural changes of the past three years has been the adoption of Mixture of Experts (MoE). The idea is simple: instead of a single large feed-forward network in each layer, you have many smaller ones — "experts" — and a learned routing function that sends each token to a small subset (typically two). A model can have hundreds of billions of total parameters but activate only a fraction of them per token, keeping computation manageable.

Google's Switch Transformer in 2021 showed this could work at scale. DeepSeek's V2 and V3 models refined the approach with fine-grained expert segmentation, producing models that match or exceed dense alternatives at a fraction of the training cost. Mixtral 8x7B — an open model from Mistral AI — demonstrated that MoE could match GPT-3.5 performance while activating only about 13 billion parameters per forward pass.

3. Large Language Models: What Scaled Up

3.1 Scaling laws and emergent behavior

In 2020, a team at OpenAI published a paper documenting something unexpected: language model loss follows clean power-law relationships with three variables — compute, data, and parameters. Double any one of them and you get a predictable improvement. The implication was that scaling was not just a brute-force approach; it was a systematic strategy with predictable returns.

GPT-3, released the same year with 175 billion parameters, confirmed the implications of those laws in a striking way. Without any task-specific training, it could write coherent essays, answer trivia, translate between languages, and do simple arithmetic — simply by being asked in plain language. Researchers called these "emergent" capabilities because they were not explicitly trained for and appeared to arise from scale alone. Whether that framing is exactly right is still debated, but the practical reality was hard to dispute: the model could do things smaller models could not.

ChatGPT (November 2022) combined GPT-3.5 with Reinforcement Learning from Human Feedback — a technique where human annotators rank model outputs and a reward model is trained on those rankings, which then guides further model fine-tuning via reinforcement learning. The result was a model that was dramatically more helpful and less likely to say harmful things than a base pretrained model. The public response was, by any measure, extraordinary: a million users in five days.

3.2 The open-source response

Meta's release of LLaMA in February 2023 changed the competitive landscape. LLaMA was a family of smaller, well-trained models — 7B to 65B parameters — that outperformed GPT-3 on many benchmarks despite being far smaller. The key insight was that most large models at the time were undertrained: they could be trained for longer on more data and get significantly better results. LLaMA's release, combined with community fine-tuning, spawned dozens of capable variants within weeks.

LLaMA-2 and LLaMA-3 followed, with LLaMA-3.3 70B in 2024 performing competitively with proprietary models on standard benchmarks. Mistral AI, a French startup, released Mistral 7B which outperformed LLaMA-2 13B despite having half the parameters — demonstrating that architectural choices (sliding window attention, GQA) could substitute for raw scale.

DeepSeek, a Chinese lab, became arguably the most significant open-source story of 2024–2025. DeepSeek-V3 trained a 671B parameter MoE model at a reported cost of around six million dollars — a fraction of what comparable proprietary training runs cost. DeepSeek-R1 then showed competitive performance with OpenAI's o1

reasoning model using a training approach (Group Relative Policy Optimization) that does not require a separate reward model. Both models were released openly, triggering a significant reassessment of assumptions about the cost structure of frontier AI.

3.3 Getting models to do what you want

A pre-trained language model is not naturally a useful assistant. It has learned to predict likely next tokens, which means it will happily continue any prompt — including harmful or incoherent ones — if that seems statistically plausible. The field of alignment research addresses how to change this.

RLHF (described above) was the first widely deployed solution and is still used by OpenAI and others. Direct Preference Optimization (DPO), introduced in 2023, simplifies this considerably: instead of training a separate reward model, you directly optimize the language model on human preference pairs. It is more stable and cheaper, and has been widely adopted in open-source fine-tuning. Constitutional AI, Anthropic's approach, goes further: a set of principles is used to generate AI feedback, which then guides training — reducing reliance on large-scale human annotation while still producing more controllable behavior.

4. Beyond Text

4.1 Images

The Vision Transformer (ViT), released by Google Brain in 2020, took the same architecture used for language and applied it to images. An image is divided into fixed-size patches, each patch is flattened into a vector, and those vectors are processed as a sequence. Pretrained on large enough datasets, ViT matched and then exceeded the accuracy of convolutional networks on ImageNet classification — which had been the benchmark for image models for a decade.

CLIP (2021) trained a vision Transformer alongside a text Transformer to jointly understand images and descriptions, enabling zero-shot image classification. DALL-E used a Transformer to generate images from text descriptions. Stable Diffusion and its successors combined diffusion models with Transformer-based text encoders. By 2024, Transformer-based text-to-image and text-to-video generation was producing results that were difficult to distinguish from photographs and film at first glance.

4.2 Protein structure: AlphaFold

AlphaFold2 is probably the most consequential scientific application of Transformer technology so far. The protein folding problem — predicting the three-dimensional structure of a protein from its amino acid sequence — had resisted solution for fifty years. AlphaFold2 solved it.

The architecture is novel: an "Evoformer" that processes both the sequence of the protein and pairwise relationships between all residues simultaneously, using a form of attention that updates both representations iteratively. The predicted structures match X-ray crystallography results to within the margin of experimental error. DeepMind subsequently released predicted structures for over 200 million proteins — essentially the entire known proteome — for free. That database is now used routinely in drug discovery pipelines worldwide.

> **Why this matters for medicine**
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> Before AlphaFold, determining a protein's structure took months to years of lab work. Drug development depends critically on protein structure — you need to know the shape of a target to design a molecule that binds to it. AlphaFold compressed what used to be a significant experimental bottleneck into a database lookup. The practical effects are beginning to show up in drug discovery timelines.

4.3 Scientific and climate applications

Transformer-based weather models have overtaken traditional numerical weather prediction for medium-range forecasting. Google DeepMind's GraphCast and Huawei's Pangu-Weather both outperform the European Centre for Medium-Range Weather Forecasts on 10-day predictions — the gold standard for operational meteorology — while running orders of magnitude faster. A ten-day forecast that used to take a room full of supercomputers now takes minutes on a single GPU.

In energy systems, Transformers are being applied to smart grid management, electricity load forecasting, and fault detection. Their ability to model complex temporal dependencies across large numbers of sensors makes them well-suited for these tasks, where traditional models struggled with non-stationarity and rare events.

5. Transformers in Drug Discovery

5.1 The pipeline and where AI fits

Developing a new drug typically takes 10–15 years and costs over a billion dollars. Most of that cost comes from failure — the vast majority of drug candidates that enter clinical trials do not make it to approval. AI is being applied at almost every stage of this pipeline in an effort to reduce those failure rates.

The earliest stages — identifying which biological target to go after, and finding molecules that might affect it — are where Transformer models have had the most impact so far. Biomedical language models like BioBERT and BioGPT, pre-trained on millions of scientific papers, can extract relationships between genes, proteins, and diseases from the literature at a scale no human team could match. PubMedBERT, trained specifically on biomedical text, reached state of the art on named entity recognition and question-answering tasks relevant to target identification.

5.2 Designing molecules

Molecular generation is a natural fit for language modeling. SMILES notation — a way of representing molecular structure as a string of characters — allows molecules to be treated as sequences. A Transformer trained on large databases of known molecules learns the grammar of chemical space and can generate novel compounds with specified properties: high binding affinity for a target protein, good solubility, low toxicity.

Multimodal approaches combine this sequence-level generation with 3D structural information from AlphaFold, allowing the model to generate candidates shaped for a specific binding pocket. Early results suggest these approaches can substantially reduce the number of experimental iterations needed before a viable lead compound is identified.

> **Case in point: Insilico Medicine**
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> In 2023, Insilico Medicine reported positive Phase IIa trial results for ISM001-055, a drug candidate for idiopathic pulmonary fibrosis — a serious and poorly-treated lung condition. The compound was identified using an AI pipeline combining Transformer-based target identification, generative molecular design, and predictive ADMET modeling. It is widely considered the first AI-designed drug candidate to reach Phase II clinical trials, and its results represented a meaningful proof of concept for the entire field.

5.3 Industry uptake

By early 2024, all twenty of the world's largest pharmaceutical companies had integrated AI into at least part of their R&D pipeline. The wave of partnerships between established pharma and AI-native biotechs accelerated in 2024–2025: Sanofi with OpenAI and Formation Bio, Roche with Recursion Pharmaceuticals, Bayer with Recursion as well. The Recursion-Exscientia merger in 2024 combined high-throughput phenomic screening with automated molecular design in a single company, signaling consolidation of the AI drug discovery sector.

SandboxAQ, spun out of Google and backed by NVIDIA, released a public dataset of 5.2 million synthetic 3D molecular structures in 2025 to give AI models better training data for predicting protein-drug binding at early development stages. The gesture reflected a broader trend toward open-science approaches in a field that has historically been extremely proprietary.

6. The Problem with Attention at Scale

6.1 Quadratic complexity

Self-attention compares every token to every other token. With N tokens in a sequence, that is N^2 comparisons. For a thousand tokens: a million. For a hundred thousand tokens: ten billion. Time and memory both scale quadratically with sequence length, which is manageable for text paragraphs but increasingly painful for long documents, entire codebases, multi-hour audio recordings, or video.

Flash Attention (2022) significantly reduced the memory footprint by reordering operations to stay within fast on-chip SRAM rather than repeatedly reading and writing to main GPU memory. Flash Attention 2 and 3 refined this further. These improvements made training on long sequences practical but did not change the underlying $O(N^2)$ computation — they just made the constant factor much smaller.

6.2 Mamba and the SSM alternative

State Space Models (SSMs) offer a fundamentally different approach. Rooted in control theory, they model sequences as signals evolving through a latent state described by differential equations. The key property is linear complexity: O(N) time and memory regardless of sequence length. The Structured State Space model (S4) in 2022 demonstrated that SSMs could handle very long-range dependencies — sequences of tens of thousands of steps — that LSTMs and even Transformers struggled with.

The limitation of S4 was that its state transitions were fixed during inference, meaning the model could not adjust what it remembered based on content. Mamba (2023) fixed this with Selective State Space Models: input-dependent transitions that allow the model to filter irrelevant information and selectively retain what matters. On language benchmarks, Mamba matched Transformer performance at equivalent parameter counts while being significantly faster at inference for long sequences.

Mamba-2 (2024) went further by showing mathematically that SSMs and attention are related operations — the SSM recurrence can be reformulated as a structured matrix multiplication, which means it can use the same GPU tensor cores as standard attention. This makes Mamba-2 nearly as fast to train as a Transformer while retaining linear-time inference.

Architecture	Complexity / Properties
Standard Transformer	$O(N^2)$ time and memory — straightforward but expensive at long context
Flash Attention	$O(N^2)$ compute, O(N) memory — same math, better

Architecture	Complexity / Properties
	GPU utilization
Mamba (SSM)	O(N) time and memory — input-selective state; matches Transformer on language
Mamba-2 (SSD)	O(N) inference — unifies SSM and attention mathematically; Transformer-speed training
xLSTM-7B	Linear complexity — 3.5x faster training than equivalent Transformer
Griffin (Google)	Hybrid: gated linear recurrence + local attention windows; strong efficiency trade-off
RetNet (Microsoft)	Parallel training + recurrent inference — designed for on-device deployment
LongNet	Dilated attention — O(N) effective; extends context to theoretically 1B+ tokens

Table 2 — Architectural alternatives to full quadratic attention and their properties

6.3 Hybrid architectures

The cleanest finding from this literature is that you probably do not need full attention everywhere. The H3 model used only two attention layers in a 24-layer network and matched Transformer performance. Meta's BiGS replaced all attention in BERT with gated SSMs and got comparable results. Google's Griffin mixes gated linear recurrences with local attention windows.

The emerging consensus is that attention is most valuable for tasks requiring the model to relate distant parts of a sequence — complex reasoning, multi-document synthesis, structured generation. For the routine work of processing sequential information, SSM layers are cheaper and often equally effective. Future architectures will likely use both, placing attention strategically rather than at every layer.

7. Agentic Systems and Multimodal Models

7.1 From chat to action

Until recently, most deployed language models operated in a simple loop: a user sends a message, the model replies. The "agentic" paradigm breaks that constraint. An agent perceives its environment — through text, images, tool outputs — plans sequences of actions, calls external tools (web search, code execution, APIs, databases), and revises its plan based on what it learns. The model is no longer a chat interface; it is something closer to an automated worker.

By 2025, all major frontier labs were investing heavily in agentic capabilities. OpenAI's Operator can interact with web interfaces. Anthropic's Claude demonstrates multi-step computer use. AI Scientist (Sakana AI) can autonomously generate research pipelines and draft papers — though with significant human oversight still required. NVIDIA's NitroGen achieved around 52% higher task success rates on novel environments across more than a thousand video game scenarios, suggesting that models are beginning to generalize reasoning strategies to unfamiliar settings.

7.2 Multimodal models

The most capable current models process text, images, audio, and video within a single context window. GPT-4o handles real-time audio and vision. Google's Gemini 1.5 Pro supports a one-million token context that can hold hours of video or thousands of pages of documents. Claude 3 Opus and Sonnet reason across image and text jointly, not as separate modules.

In healthcare specifically, Google's Gemini 3 (unveiled at Google I/O 2025) includes purpose-built variants for radiology image interpretation, clinical documentation, and real-time decision support with electronic health record integration. These are not general-purpose models applied to medicine; they are fine-tuned on clinical data with evaluation frameworks designed for medical accuracy.

7.3 Context windows and what they make possible

GPT-4 launched with a 32,000 token context window — about 24,000 words. By 2025, frontier models routinely supported one million tokens or more. LongNet, a Microsoft Research architecture using dilated attention, demonstrated contexts of up to one billion tokens in theory. In practice, the question has shifted from "can the model hold a long context?" to "can it reason across one?" — a harder problem that is still not fully solved.

Long contexts enable a different class of applications: analyzing entire software repositories for bugs, synthesizing thousands of papers on a research question,

maintaining coherence across a book-length document, or processing a full day's sensor data from a medical device. These are not incremental improvements on shorter-context tasks; they are qualitatively different capabilities.

8. Limitations Worth Taking Seriously

8.1 Hallucination

Language models generate plausible-sounding text. That is what they were trained to do. When a model does not know something, it often produces a confident-sounding answer anyway — fabricating citations, inventing facts, misremembering details. This is called hallucination, and it is not a bug that will be fixed in the next version. It is a consequence of the training objective.

Mitigation strategies exist. Retrieval-augmented generation (RAG) grounds responses in retrieved documents, giving the model a factual basis to work from. Tool use allows models to look things up rather than relying on internal knowledge. Better training on factual accuracy using RLHF reduces the frequency of hallucinations. But none of these eliminate the problem, and in high-stakes domains — medicine, law, engineering — a model that is mostly right is not good enough.

8.2 Bias

Training on internet-scale text means training on a document collection that reflects the prejudices, blind spots, and imbalances of whoever produced that text. Large language models can and do generate outputs that are discriminatory across gender, race, nationality, religion, and disability. This is not a hypothetical concern; it has been documented extensively. The difficulty is that the biases are diffuse and context-dependent, making them hard to characterize comprehensively or to correct systematically.

8.3 Interpretability

We do not have a good account of what large Transformers are doing internally. Mechanistic interpretability research has identified specific attention heads that appear to track subject-verb agreement, or copy tokens, or detect certain syntactic structures. But these are pieces of a puzzle that is nowhere near complete. For a model with hundreds of billions of parameters, we understand only a tiny fraction of what is happening in any forward pass.

This matters practically. In regulated industries, decisions that affect people's lives must often be explainable. A model that cannot be audited in a meaningful way is hard to deploy in healthcare, finance, or legal contexts. And from a safety perspective, a system you cannot inspect is one whose failure modes you cannot anticipate.

8.4 Computational and environmental cost

Training frontier models requires substantial energy. GPT-3's training run has been estimated at roughly the carbon footprint equivalent of several international flights. Training runs for 2024–2025 frontier models are considerably larger. Inference at global scale — hundreds of millions of queries per day — adds ongoing costs. The field is improving on efficiency: MoE architectures, quantization, smaller but better-trained models, and distillation all reduce the cost per query. But as capability and usage both grow, the net environmental footprint continues to increase.

9. Regulation and What Comes Next

9.1 The regulatory picture

The EU's AI Act entered into force in August 2024, establishing the most comprehensive AI regulatory framework yet. It is risk-based: AI systems are categorized by the harm they could cause, with the most stringent requirements falling on "high-risk" applications in healthcare, education, law enforcement, and critical infrastructure. Providers of general-purpose AI models above certain capability thresholds face additional transparency and evaluation requirements.

The US has taken a different approach — primarily through executive orders and NIST guidelines rather than statute. The FDA and EMA have both issued early guidance on AI in drug development, focusing on validation, transparency, and accountability. These frameworks are still maturing; the regulatory environment for AI in medicine in particular is likely to look substantially different by 2027 than it does today.

The question of open-source models sits uncomfortably within existing frameworks. Open releases democratize access and accelerate research, but they also make capability restrictions largely unenforceable once a model is published. This tension has not been resolved, and significant policy disagreement remains about where to draw the line.

9.2 Where the research is heading

A few themes appear to be organizing the next wave of work.

Reasoning. Current models are surprisingly good at surface-level tasks but still fail on problems requiring careful, step-by-step formal reasoning — mathematics, logic, planning. OpenAI's o1 and o3 series, and DeepSeek-R1, use extended chain-of-thought reasoning during inference — essentially giving the model time to think — with significant improvements on hard benchmarks. This direction is being pursued aggressively.

Persistent memory. Models have no memory between conversations. Every session starts from zero. Solving this — allowing models to learn continuously from new information without expensive retraining — is one of the harder open problems in the field. Hybrid SSM-attention architectures, external memory stores, and retrieval augmentation are all partial solutions.

Efficiency at the frontier. DeepSeek's training cost revelations in early 2025 reset assumptions about how much compute is necessary to train frontier models. Algorithmic improvements in data selection, curriculum, and training dynamics

appear to have more headroom than the field had assumed. Efficient training is now a primary research priority, not an afterthought.

Interpretability as a safety tool. Anthropic, DeepMind, and academic groups are investing substantially in mechanistic interpretability — understanding, at a circuit level, what computations trained networks are performing. This is motivated partly by scientific curiosity and partly by the recognition that as AI systems are given more autonomy, the ability to audit their reasoning becomes a safety requirement, not just a nice-to-have.

10. Conclusion

The Transformer has proven to be one of the most generative architectural ideas in the history of machine learning. It is not one invention but a design space — a set of principles that has been applied, extended, and refined across nearly a decade by thousands of researchers. The 2017 paper described a mechanism for machine translation. The architecture that mechanism spawned now underlies systems that can predict protein structures, forecast the weather, design drug molecules, read medical images, and generate coherent video from text descriptions.

The open questions are real. Transformers are expensive to run, prone to hallucination, and largely opaque in their reasoning. The quadratic complexity of attention imposes fundamental constraints that alternative architectures are beginning to address in earnest. Regulatory frameworks for these systems are only beginning to take shape.

But the core trajectory is clear: models are becoming more capable, more efficient, and more deeply embedded in consequential systems. Understanding what they are, how they work, and where they fall short is no longer optional background knowledge for researchers and practitioners in the relevant fields. It is an operational requirement.

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